



# SUJET STATAPP 323 - 2026-2027

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**Organisme :** CREST

**Catégorie :** Santé

**Sujet :**

**How can we measure health and physical dysregulation? A multiverse analysis of The Irish Longitudinal Study on Aging**

## Description du sujet :

Population health research increasingly relies on high-quality biomarker data from large, representative samples: survey participants don't just answer questions about their health, they also see a nurse who measures their height and blood pressure and takes a blood sample, so that health markers like inflammation or cholesterol can be measured.

One particularly promising construct built from a large number of biomarkers is allostatic load, a measure of physical dysregulation commonly described as the "wear and tear" the human body experiences. Allostatic load is theoretically well understood?it generally behaves, and affects other variables, the way scientists expect?but there is no consensus on how it should be measured. Measures are often based simply on which biomarkers happen to be available in a given data set, and the consequences of including or excluding a biomarker are poorly understood. Researchers also make different choices about how to treat each biomarker: sometimes they split a measure into "healthy" and "unhealthy" categories using a clinical cutoff (for instance, treating a BMI under 25 as healthy and anything above as not); sometimes they use the full range of a measure instead (treating a BMI of 26 as healthier than a BMI of 28); and sometimes they adjust for medication (someone whose blood pressure is in the healthy range only because they take medication for it might still be counted as having high blood pressure).

This leaves researchers with substantial flexibility in how they calculate allostatic load for their analyses. This project uses specification curve analysis?a type of "multiverse analysis" that runs many plausible versions of the calculation side by side?to see how sensitive well-established associations, such as those between allostatic load and age or education, are to these different calculation choices.

Statistical software will be Stata or R, depending on the students' preferences.

At the end of this project, students will have:

- An excellent understanding of specification curve analysis, a flexible tool for understanding how different model specifications can affect the results of an analysis. This tool is useful, in principle, for any analysis that leaves the researcher a lot of room for decision-making.
- An excellent understanding of The Irish Longitudinal Study on Aging (TILDA). Since TILDA belongs to a large family of similar aging studies from many countries (<https://g2aging.org/>), understanding it will give you a head start in using any of the others, such as the US Health and Retirement Study or the Survey of Health, Aging and Retirement in Europe (which also includes France).
- With effort, perseverance, and a bit of luck, the chance to publish an article in an international peer-reviewed journal.

## Données du sujet :

The Irish Longitudinal Study on Aging (TILDA) is a prospective cohort study designed to provide an evidence base for addressing the challenges and opportunities associated with population aging in Ireland. It's a nationally representative study of community-dwelling older persons aged 50 years or more and their spouses living in the Republic of Ireland.

**Modèles envisagés du sujet :**

Specification curve analysis (a type of multiverse analysis) to test how sensitive associations between allostatic load and variables like age or education are to different ways of calculating allostatic load (e.g., which biomarkers are included, clinical cutoffs vs. continuous measures, adjustment for medication use). Statistical software: Stata or R, depending on student preference.

**Bibliographie du sujet :**

Kearney, Patricia M., Hilary Cronin, Claire O'Regan, Yumiko Kamiya, George M. Savva, Brendan Whelan, and Roseanne Kenny. 2011. "Cohort Profile: The Irish Longitudinal Study on Ageing." *International Journal of Epidemiology* 40(4):877-884. doi: <https://doi.org/10.1093/ije/dyr116>

Donoghue, Orna A., Christine A. McGarrigle, Margaret Foley, Andrew Fagan, James Meaney, and Rose Anne Kenny. 2018. "Cohort Profile Update: The Irish Longitudinal Study on Ageing (TILDA)." *International Journal of Epidemiology* 47(5):1398-1398. doi: <https://doi.org/10.1093/ije/dyy163>

These two articles include generic information about TILDA.

Simonsohn, Uri, Joseph P. Simmons, and Leif D. Nelson. 2020. "Specification Curve Analysis." *Nature Human Behaviour* 4(11):1208-1214. doi: <https://doi.org/10.1038/s41562-020-0912-z>

This article is a pretty good introduction into specification curve analysis.

McLoughlin, Sinéad, Patrick Präg, Mel Bartley, Rose Anne Kenny, and Cathal McCrory. 2022. "Intergenerational Social Mobility and Allostatic Load in Mid-Life and Older Ages. A Diagonal Reference Modeling Approach." *Journals of Gerontology: Series B*:1-13. doi: <https://doi.org/10.1093/geronb/gbac122>

This is allostatic load research using TILDA data in action.

**Notation du sujet :****Commentaires :**